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SECTION 1

PHASE 3: Visualization & Dashboard

PROJECT TITLE:

**Business Intelligence Analysis of Brazilian E-Commerce Customer Purchasing Behavior
Using Olist Datasets**

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1.0 Dashboard Overview

This analysis presents two comprehensive Power BI dashboards with interactive visualizations and storyboards designed to provide actionable insights into sales performance, product category analysis, logistics efficiency and geographic distribution of operations.

Dashboard 1: Sales & Product Performance Dashboard

The main dashboard displays comprehensive KPI metrics and visualizations of sales trends, product performance, geographic distribution, and customer behavior.

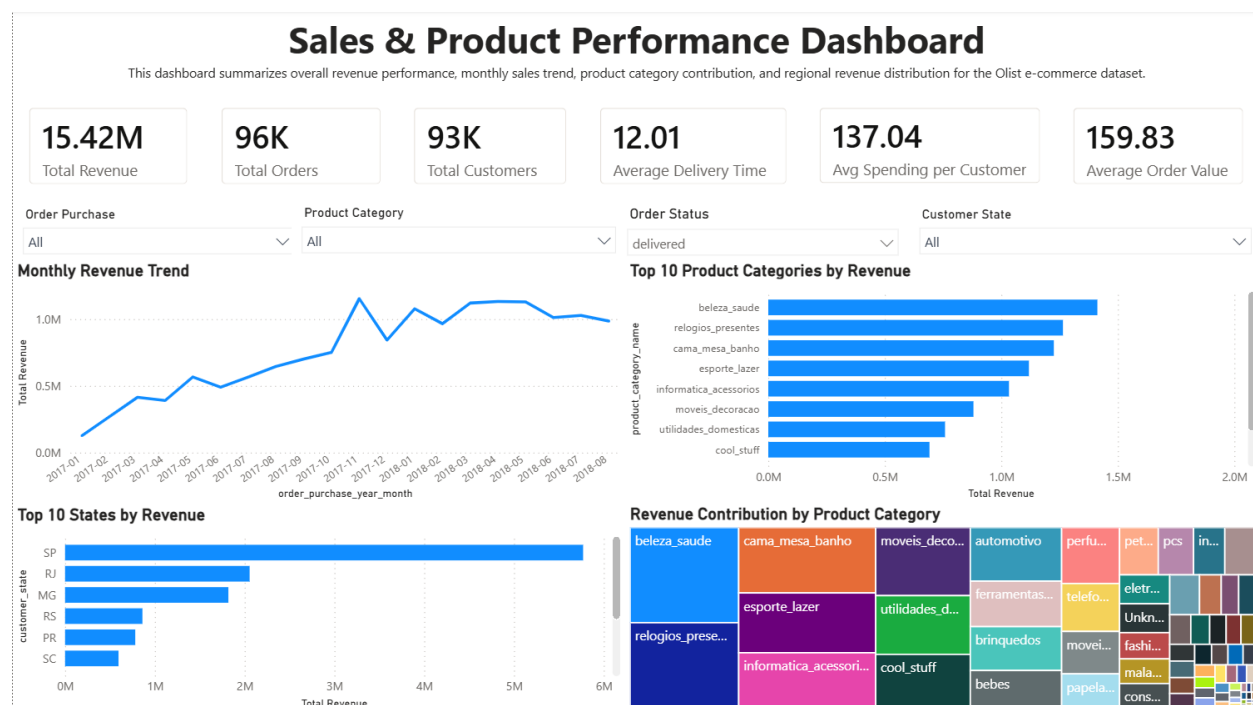


Figure 1: Sales & Product Performance Dashboard

Key Metrics Displayed:

- Total Revenue: \$15.42M
- Total Orders: 96K
- Total Customers: 93K
- Average Delivery Time: 12.01 days
- Avg Spending per Customer: \$137.04
- Average Order Value: \$159.83

Dashboard 2: Logistics & Geography Dashboard

The logistics dashboard provides an overview of order fulfillment, delivery status, geographic customer distribution, and state-level performance metrics.

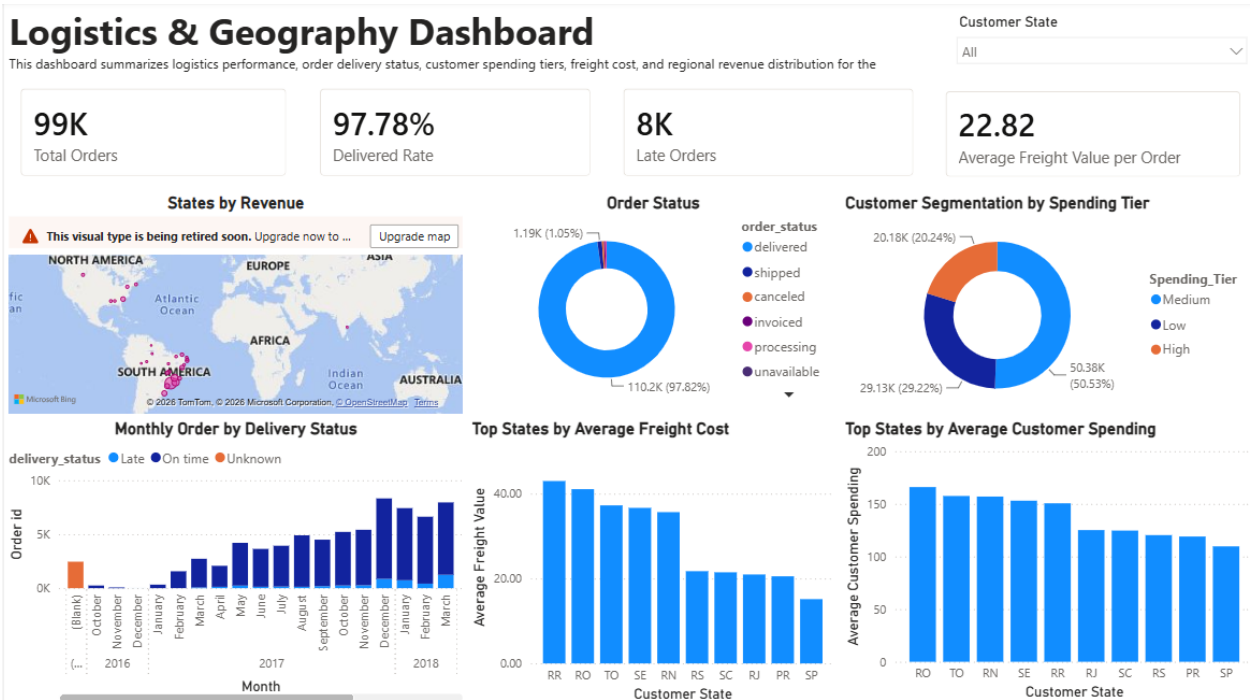


Figure 2: Logistics & Geography Dashboard

Key Metrics Displayed:

- Total Orders: 99K
- Delivered Rate: 97.78%
- Late Orders: 8K
- Average Freight Value per Order: \$22.82

2.0 Key Insight

2.1 Revenue Performance & Growth Trends

The organization has achieved a total revenue of \$15.42M across 96,000 orders from 93,000 customers. Highest revenue in November at approximately \$ 1.15M, driven by seasonal shopping demand. The lowest revenue was recorded in January 2017 at approximately \$127.4K, indicating a strong year-end purchasing effect. The average order value of \$159.83 demonstrates stable customer purchasing power and efficient order management.

2.2 Product Category Performance & Market Concentration

The Top 10 Product Categories analysis shows that 'beleza_saude' (Beauty & Health) which contribute about \$1.4M is the leading revenue generator, followed by 'religios_presentes' (Religious Gifts) with \$1.26M and 'cama_mesa_banho' (Bedding & Bath) with \$1.23M. This indicates significant market concentration in lifestyle and wellness categories. The product portfolio demonstrates diversification across multiple categories, reducing business risk while allowing for target marketing strategies.

2.3 Geographic Revenue Distribution & Regional Market Strength

The geographic analysis reveals that Sao Paulo (SP) is the dominant market, generating the highest revenue, followed by Rio de Janeiro (RJ) and Minas Gerais (MG). This concentration in major metropolitan areas suggests strong market penetration in densely populated regions. The presence of revenue across multiple states indicates geographic diversification, presenting opportunities for expansion into underperforming regions.

2.4 Delivery Performance & Logistics Efficiency

The logistics dashboard shows an impressive delivered rate of 97.78%, indicating excellent order fulfillment capability. However, 8,000 late orders represent a notable concern requiring attention. The average delivery time of 12.01 days suggests a reasonable fulfillment window, though optimization opportunities exist. The average freight value per order of \$22.82 indicates moderate logistics costs, but regional variations suggest potential for cost optimization in higher-cost regions.

2.5 Customer Segmentation & Spending Behavior

The customer segmentation analysis reveals three distinct spending tiers: High-value customers (56,444 customers, 50.1%), Medium-tier customers (34,644 customers, 30.75%), and Low-value customers (21,576 customers, 19.15%). This distribution indicates healthy customer base composition with significantly high-value customer concentration. The average customer spending of \$137.04 reflects solid customer lifetime value.

3.0 Business Recommendations

3.1 Optimize Delivery Performance & Reduce Late Orders

While the 97.78% delivery rate is commendable, the 8,000 late orders impact on customer satisfaction and operational efficiency. Recommendations:

- Conduct root cause analysis on late orders to identify bottlenecks
- Implement real-time tracking and proactive customer notification systems
- Negotiate with logistics partners to reduce delivery times to under 10 days
- Target: Reduce late orders to below 5% within 6 months

3.2 Expand High-Performing Product Categories

Leverage the strong performance of beauty & health and religious gifts categories:

- Increase inventory in top 5 product categories contributing 60% of revenue
- Launch seasonal campaigns targeting peak demand periods
- Develop cross-selling strategies between complementary categories
- Target: Increase average order value by 15% annually

3.3 Geographic Market Expansion & Regional Development

Develop strategies to penetrate underperforming regions:

- Analyze regional demand patterns and tailor product offerings
- Establish regional distribution centers in high-demand states
- Implement targeted digital marketing campaigns for emerging markets
- Target: Achieve 25% revenue growth from secondary markets within 12 months

3.4 Optimize Logistics Costs & Regional Performance

Address regional freight cost variations:

- Analyze high-freight-cost states and negotiate better rates
- Implement regional logistics hubs to optimize distribution networks
- Target: Reduce average freight cost by 10-15% through optimization

3.5 Customer Segmentation Strategy & Retention Focus

Develop tailored strategies for each customer segment:

- Implement loyalty programs targeting high-value customers
- Launch upselling campaigns to move medium-tier customers to high-value status
- Design re-engagement campaigns for low-value customers
- Target: Increase average customer spending from \$137.04 by 20%